MD. MEHEDI HASAN MOON

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EDUCATION

Shahjalal University of Science and Technology, Sylhet, Bangladesh 2025 to present
M.S. in Physics (Mixed Mode)

Shahjalal University of Science and Technology, Sylhet, Bangladesh 2019 to 2025
B.Sc. (Honours) in Physics. CGPA 3.04 out of 4.00.

Project: *Numerical solution of a boundary value problem of the Legendre differential equation for P_4 using the finite difference method.*

Supervisor: Prof. Dr. Mohammed Sujaul Haque Chowdhury

M.S. THESIS

Title: "Local-Excitation Interference, Not Donor–Acceptor Distance, Controls the Radiative Strength of a Through-Space Charge-Transfer TADF Emitter"

Abstract:

This thesis asks if exchange coupling and radiative strength respond alike. The test case is STF-DBTS, a through-space charge-transfer TADF emitter. Calculations use ωB97X-D/def2-SVP with continuum toluene solvation. Local triplets split the triplet charge-transfer configuration over twelve states. Only a diabatic gap measures the exchange splitting, +50.6 meV. It is built alike on both spin manifolds. Along the fluorene-acceptor torsion, the exchange integral varies two-fold. It tracks HOMO-LUMO overlap (r = 0.92). Interference with a small acridine-local excitation sets the S_1 transition dipole. At one conformer, the gap falls 19% but the dipole falls 45%. These results point to overlap and local-excitation admixture as design variables.

Advisor:

Prof. Dr. Nazia Chawdhury (Physics, SUST)

PUBLICATIONS

Manuscripts under review

MD. Mehedi Hasan Moon, Mohammad Afsar Uddin and Nazia Chawdhury. "Bridge-Dependent Emission Tuning and Radiative-Rate Suppression in a Donor–Acceptor [3.3]Paracyclophane Series: A Gas-Phase TD-DFT Study." *Under review, 2026.*

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PRESENTATIONS

MD. Mehedi Hasan Moon and Nazia Chawdhury. "Local-Excitation Interference, Not Donor–Acceptor Distance, Controls the Radiative Strength of a Through-Space Charge-Transfer TADF Emitter." International Conference of Physical Sciences (ICPS-2026), SUST, Sylhet, Bangladesh, December 2026. *Oral presentation, accepted.*

RESEARCH EXPERIENCE

Research Student, M.S. Thesis Mar 2026 to present
Organic Optoelectronics Lab, Department of Physics, SUST. Advisor: Prof. Dr. Nazia Chawdhury.

- Characterised STF-DBTS, a reported through-space charge-transfer TADF emitter.
- Showed the singlet-triplet gap must be built diabatically on both spin manifolds.
- Computed the HOMO-LUMO exchange integral in PySCF on a relaxed torsional scan.
- Traced the S_1 transition dipole to interference with an acridine-local excitation.

Research Assistant May 2025 to Mar 2026
Department of Physics, SUST

- Ran geometry optimisations and excited-state calculations on organic emitters.
- Assigned excited-state character with wavefunction analysis.

Research Project, Nuclear Astrophysics 2025 to present
Department of Physics, SUST. Supervisor: Anock Somadder, Associate Professor.

- Coupling the SkyNet reaction network to the MESA stellar evolution code.
- Goal: time-resolved neutrino emission from neutron-capture burning.

Undergraduate Research Project Jun 2023 to Dec 2023
Department of Physics, SUST. Supervisor: Prof. Dr. Mohammed Sujaul Haque Chowdhury.

- Solved a Legendre boundary value problem for P_4 by finite differences.

TEACHING EXPERIENCE

Supplementary Instructor, physics and mathematics

Taught secondary and post-secondary students. Designed lesson plans around practical, hands-on learning. Gave coursework feedback aimed at real improvement.

ACADEMIC SERVICE

Convener, International Conference for Advances in Physics, SUST Dec 2025

Led the organising committee for international participants. Covered logistics, scheduling and technical sessions with faculty.

PROFESSIONAL DEVELOPMENT

Participant, Research Workshop: The Path to Academic Excellence Jan 2026

SUST Research Society (SUSTRS). Research ethics, journal publishing, paper structure and Zotero.

HONORS AND AWARDS

Government Scholarship, Secondary School Certificate 2015

Awarded on Secondary School Certificate results.

Government Scholarship, Junior School Certificate 2012

Awarded on Junior School Certificate results.

RESEARCH INTERESTS

Excited states of organic emitters, especially TADF and through-space charge transfer.
Structure-property relationships in π-stacked donor-acceptor systems.
Electronic structure methods: DFT, TD-DFT and wavefunction analysis.
Organic optoelectronic devices, including OLEDs and organic photovoltaics.

TECHNICAL SKILLS

Software: Gaussian 16, ORCA 6.0, Multiwfn, PySCF
Methods: DFT, TD-DFT, MP2, CPCM solvation, NTO and hole-electron analysis
Programming: Python (Matplotlib), C++, Bash scripting, Mathematica
Tools: Linux, Git and GitHub, Zotero, Microsoft Office

LANGUAGES

Bengali (native)
English

REFERENCES

Available on request.